SUPPORT & FAQ

# MedSim VRAI — TrainingBridge (V8 on-prem)

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## GETTING STARTED

**How do I start the system on my Mac for a local run (just the operator's browser)?**

Open Terminal and run: bash scripts/run_cards.sh from the project folder. The portal boots (you'll see "Uvicorn running on https://…:8760") and Chrome opens straight to the Mission Control card UI. You'll land on the Set up tab. This local launcher binds to the Mac only (https://127.0.0.1:8760), so it's for Mac-browser testing, not tablets. Stop it with Ctrl+C in that terminal.

**How do I start the system so iPads and tablets on the Wi-Fi can reach it?**

Use the LAN launcher, not the local one: bash scripts/start.sh. This binds the whole network (not just the Mac), auto-fixes the TLS certificate for the current Wi-Fi IP, serves the patient avatar app from the one correct origin, and opens the operator console at the LAN address so every QR code carries the LAN IP (never 127.0.0.1). run_cards.sh binds loopback only, so tablets get "portal unreachable" against it — always use start.sh for tablets. After it's up, re-print the QR sheet (old codes may point at a previous IP).

**What's the difference between run_cards.sh and start.sh — which do I run?**

run_cards.sh is the local, card-first launcher for Mac-browser-only work (https://127.0.0.1:8760); it binds the loopback so tablets cannot reach it. start.sh is the LAN launcher for tablet/field sessions: it binds 0.0.0.0, re-mints the cert for the current IP, and opens the console at the LAN URL so QR codes carry the real IP. Rule of thumb: no tablets = run_cards.sh; any tablet in the room = start.sh.

**Is there a one-command check to confirm tablet pairing will work before a session?**

Yes — run bash scripts/preflight.sh before anyone scans a QR. It checks, in ~10 seconds: your Mac's LAN IP (flags an IPv6-only hotspot that can't pair), that the portal is listening and serving the one-origin app, that the TLS cert covers the current IP, that the onboarding helper is up, and it regenerates the QR for the current network. Pass a tablet IP (preflight.sh 192.168.1.169) to also ping-test that tablet. It prints PREFLIGHT PASS or lists exactly what to fix. This ritual exists so a router/IP/cert drift is caught in seconds instead of burning a session.

**How do I stop the server?**

If you launched with run_cards.sh, press Ctrl+C in that terminal. If you launched with start.sh (LAN mode, runs in the background), run bash scripts/start.sh stop. Either way you can also stop it with pkill -TERM -f run_portal.py.

**How do I sign in, and what do Admin, Instructor, and Observer mean?**

On the sign-in screen, pick your seat (Instructor, Admin, or Observer · read-only), then enter the master password to unlock. On first run you're asked to create the master password — it encrypts your saved keys, so keep it safe. The master password is the Admin seat. Instructor and Observer can each be given their own password (see Credentials); whichever password you type at sign-in decides your seat. Only the Admin seat can open the Credentials page and EHR admin. Use Admin for credential/EHR-admin work; Instructor is fine for running scenarios and network.

**After I restart the server, do I need to sign in again?**

Yes. The encrypted vault is only held in the server's memory while you're signed in, so a restart clears it and you must re-login. Signing back in also re-arms the AI (Anthropic) key so characters answer normally, and the room resumes on its own — you don't need to rebuild it. Sign in with the same seat you used before (Admin for credential checks).

**Characters won't reply and readiness says no AI key — where do I add it?**

Sign in as Admin, then open the Credentials page. Under "Anthropic API key," paste the key into the field and click Save; a "Stored" badge appears and you can click Test to confirm it's accepted. Without this key the Voice / AI readiness check is red and characters can't generate replies. The key is stored encrypted at ~/.medsim/vault.enc and only decrypted in memory while you're signed in.

## SESSIONS & SCENARIOS

**How do I set up and launch a scenario from the card UI?**

On the Set up tab, work down the Launch Wizard: choose the EHR and the Number of beds (Patients & rooms); pick a Scenario for each bed (the bed's patient follows from it); assign Scenario characters per bed and any Shared characters (a common doctor or allied-health role); give each character a face in the image picker; add Devices per bed; then review EHR, beds, cast, and device count. When the readiness pill is green (or amber), click Launch — it opens the room dashboard and mints the device QRs. Target for a cold start is under about 90 seconds. You can also launch from the Board view.

**Where do I actually run the live session once it's launched?**

Switch to the Operate tab. You get one card per bed/patient (join, device, and character counts), one per shared character (voice state + QR), the med cart(s), and the Nursing station. Each card has Open (opens in place) and Pop out (opens in a new window you can drag to a second monitor — it stays fully live). At the top is the room lifecycle bar: Start, Pause, Resume, Inject scene, Print QR, and End all (debrief).

**What do Start, Pause, Resume, and Inject scene do?**

On the Operate lifecycle bar: Start arms a configured room. Pause and Resume gate student input (Pause stops trainees acting, Resume lets them continue). Inject scene opens a modal and posts a scene event (for example "family arrives — visitor at bedside") to every bed at once. Print QR opens the station join sheet. End all (debrief) closes the session and builds the cohort debrief.

**How do I introduce a mid-scenario event to all the patients at once?**

On the Operate tab's room lifecycle bar, click Inject scene. A modal opens; choose or enter the scene, and it posts to every bed in the room simultaneously. Use it to escalate or change the situation for the whole cohort in one action.

**What's the difference between Open and Pop out on an Operate card?**

Open expands the card in place on the Operate page. Pop out opens that one card in its own window with the page chrome hidden — useful for a second monitor. A popped-out card stays fully live off the existing polls. Pop-out targets are role-correct: a patient card opens the encounter console, a shared character opens the shared station, a med cart opens the cart device UI, the nursing card opens the nursing station, and medical records opens the records page.

**I restarted mid-session — is my room gone?**

No. On boot the system restores the last session, and after you re-login the Operate tab shows the resumed room intact (scenario, beds, devices, shared cast). A "Resumed ..." note and a Resume banner may appear. If you instead launch a fresh scenario, the resumed state clears (a new launch is not a resume). A corrupt or empty snapshot boots cleanly and simply offers to configure in Setup — it won't crash.

**Where's the old classic control room if I need it?**

Every screen has "Switch to classic control room" at the top-right, which lands on the classic setup page. From inside the classic sidebar there's a "Mission Control (cards)" link to come back. The classic room is the supported fallback. A plain launch defaults to the cards; launching with MEDSIM_DEFAULT_VIEW=classic opts back to classic.

**Can I generate a new scenario from the card UI?**

Yes — the card UI has a "Generate a scenario" entry (alongside the scenario tools, near Debrief) plus a Local context option for overlaying your site's protocols. Use it to author a new scenario without dropping back to the classic room. You can also import a scenario bundle from another MedSim system (see the scenario Import question).

**How do I export a scenario to share it with another MedSim system?**

On Set up → Scenarios, each row has an Export button (next to Edit and Docs). Click it and a .medsim-scenario.json file downloads, containing the scenario, its characters, and any support documents. Hand that file to the other system's operator to import.

### How do I import a scenario bundle someone sent me?

On the Scenarios page, use the Import… button in the header, choose the .medsim-scenario.json file, and click Import. If the ids already exist, the scenario lands under a new suffixed id (your original is untouched) and identical characters read "linked." A Review & edit button opens the imported copy. Watch the post-import checklist: an avatar image and any local-context tag do not travel with the bundle, so it will warn you to reassign the face and set the local tag. A tampered file imports with a checksum-mismatch note (not a hard failure); a file that isn't a bundle is refused with a clear message and nothing lands.

## CHARACTERS & IMAGES

### How do I add a new character?

Open the Characters page and click "+ New character." Fill in Basics (Name, Role, Identity), then the Voice section (Register, Sentence length, and 3–5 example utterances that make the voice land). Save. Character cards are stored as YAML under characters/. From a character card you can also Edit, Duplicate, Delete, or "Develop avatar" to build its animated face.

### How do I give a character a face/portrait?

In the Launch Wizard each character row has an image swatch strip — click a face to assign it. That one image is used both as the flat audio portrait and as the source photo for the 3D animated rig, so you only pick once. If there are no images yet, use the "+ image" link to upload a portrait on the Skins page, then reopen the wizard and pick it.

### There are no faces to choose — how do I upload one?

From the character image strip in the wizard, click the "+ image" (upload) link, which opens the Skins page. Upload a portrait image there, then return to the Launch Wizard and pick it from the swatch strip. The same uploaded image serves as both the flat portrait and the animated-rig source.

### What's the difference between Scenario characters and Shared characters?

Scenario characters belong to a specific bed (the patient is locked to that bed; a recurring title across beds shows · V1 / · V2 so you can tell them apart). Shared characters are a separate picker for cast common to the whole room — a common doctor or allied-health role that any bed can reach from one shared station/QR. Assign scenario characters per bed and shared characters once for the room.

## AUDIO & SPEECH

### How do I pick the voice a character speaks in?

Open a patient's card in Operate to reach the encounter console, then in the Characters · voices section pick an ElevenLabs voice per character; the ▶ Test button previews it. Push-to-talk and typed replies then speak in that assigned voice (for example Callum for Mr. Hayes), not the browser default. If no ElevenLabs key is set, the system falls back to browser text-to-speech.

### How do I role-play a character myself (talk as the doctor, etc.)?

In the encounter console's Characters · voices section, click Engage on the character — it opens an in-console chat so you can play that character live. On a shared character you Engage from the shared station. Push-to-talk and the type box both work while engaged, and replies speak in the character's assigned voice.

### Push-to-talk isn't capturing anything — the mic never activates. What do I do?

The browser needs microphone permission. Click a character chip, or focus the type box, or hold the talk button — that user action triggers the browser's microphone prompt; click Allow. If you previously blocked it, click the little site icon at the left of the address bar, set Microphone to Allow, then reload the page. The on-screen status will read "Microphone ready — hold Hold to talk and speak" once it's granted. As a fallback you can always type your line and press Enter.

### The start or end of what I say keeps getting cut off. How do I fix it?

Hold the button down for the whole utterance and don't release until just after your last word — the speech recognizer lags real speech slightly, so releasing on the final word clips the tail. Press and hold, speak, then release a beat late. On a quick, short utterance the system still captures the interim transcript and fires the turn, so fast presses aren't lost. If the ending still clips on a tablet, confirm the tablet is on the fresh build (see the stale-page fix).

### I get "No speech heard" or "Didn't catch that" even though I spoke — why?

The recognizer didn't pick up audio. Hold the button down while you're actually speaking (start talking after the hold begins, not before), speak clearly, and release a moment after the last word. If it keeps happening, the mic may be blocked — click the site icon at the left of the address bar and set Microphone to Allow, then reload. You can always fall back to typing the line and pressing Enter (this also works in Safari, where speech recognition may not).

### The mic or speech-to-text won't work in my browser at all — can I still drive a character?

Yes. Type your line in the box below the talk button and press Enter — the character replies in its assigned voice. This type fallback works everywhere the mic or speech-to-text does not, notably Safari. It's the reliable path when a browser blocks speech recognition.

### The Operate status shows Speech-to-text as cold/amber — what do I do?

Open System status on the Operate tab and, on the Speech tile, tap "Warm speech model." On the next poll it turns green. The room speech-to-text only matters for audio stations, so a cold model is amber (a nudge to warm it), never a blocker. "Test all" re-runs every readiness check; the restart hint just shows the command — it does not restart the portal for you.

### A character's action in asterisks appears in the transcript but isn't spoken — is that a bug?

No, that's intended. Text a character emits inside *asterisks* is treated as a stage direction: it's shown in the transcript so trainees can read it, but it's never spoken aloud. Only the character's actual dialogue is voiced.

### After a restart the character just echoes "I heard you say..." instead of answering — how do I fix it?

That echo means the AI (Anthropic) key isn't armed. Simply re-login as Admin — signing in re-arms the process's Anthropic key, and the next push-to-talk answers in character without restarting the room. If it still echoes after re-login, the key itself is stale or rotated: open the Credentials page and update the Anthropic key (Save, then Test). A live turn will report "Anthropic API key is invalid or rotated. Update it at /portal/credentials — the next PTT will pick up the new key without restarting the scenario."

### How does the nurse use the live intercom to talk to a bed or all beds?

On the Nursing station, each bed card has a 🎙️ Hold to talk button (live two-way audio to that bed), and the header has a 📢 Page all rooms button to reach every bed at once; there's also a typed-page box per bed as a fallback. When a bed calls back, a banner and the bed card's pip light up. On the bedside, the Integrated Com & Alarm device's Intercom tile is hold-to-talk, and the chat station has a 📞 Call nurses station button. Live audio needs HTTPS and a mic grant on first use (tap to allow).

### The nurses station stopped answering a bed after someone reloaded a page — why?

Reloading churns the live-audio peer IDs. When you reload a surface, reload EVERY surface in that room — the bedside AND the nurses station — not just the bed. The station page in particular must be reloaded to pick up the current build and re-establish its connection; reloading only the beds is not enough. For triage you can add ?icdiag=1 to the URL to show a hidden connection-diagnostic overlay.

## DEVICES & RECORDS

### How do I add bedside devices like IV pumps to a bed?

In the wizard's Devices step (or the Operate devices card / modal), use the per-bed picker and "+ Add device (mint QR)." Bed-level device kinds are IV pump (Alaris) and enteral pump (Kangaroo Omni). Each minted device gets a QR you print and scan at the bedside. Assign the device to a character/bed, then Save. Devices are minted at launch and their QRs are ready on the room dashboard.

### I can't add a med cart from a patient's device modal — the cabinet option is gone. Why?

Med carts (cabinets) are a room-level resource, so the per-patient device manager deliberately hides the Cabinet kind and only offers bed-level devices (pumps). The help text reads "Med carts are managed at the room level." Create carts at the Multi-Patient Control level and link the beds to them, rather than minting a cart inside a single encounter. (If a request slips through, the server refuses an encounter-scoped cabinet with a message pointing you to Multi-Patient Control.)

### How does a med cart show medications across multiple patients for the nurse?

A med cart is a shared, room-level dispensing resource that appears on the Common page of the QR sheet — it's the drawer holding every patient's MAR, not repeated per bed. Open the cart device (Med cart card in Operate, or the Common devices QR) to reach the cart UI. Bedside pumps stay bed-level; only carts are shared at the room level and linked to the beds you choose.

### How do I sign in to the medical records (EHR) to open a patient chart?

On the Sign in to Helix Health screen, enter a Display name and the nurse's Initials, then click "Enter records →" — the chart list is scoped to that nurse's assigned patients, and entries are tagged to whoever signed in. To see every chart instead, click "Supervisor view — all patients." Use "Switch user" to change who you're signed in as. Patients only appear here once a session is running.

### Records says "No patients in your scope" — what happened?

The initials you signed in with aren't assigned any patients in this room. Either use "Switch user" and sign in as a nurse who has assignments, or open "Supervisor view — all patients" to see every chart. Also confirm a session is actually running — patients only populate the records list once a room is live.

### Can I limit which patients a nurse sees in records?

Yes — records honor nurse assignments. When a nurse signs in with their initials, the chart list is scoped to just their assigned patients; the supervisor view shows all. On the Set up side there's a "Restrict by assignment" option controlling medical-records scope at launch, so students only see the charts for the beds they're responsible for.

### Can a student play one of the AI character roles (doctor, RT, pharmacist) instead of the AI?

Yes. Go to Set up → Assignments (or the classic roster). The role selector offers Doctor / Respiratory therapist / Pharmacist (fills the AI seat). Add a student to one, and on Operate → Network & devices that character node shows a student-colored dashed frame with the student's name. The seat reads idle until the student actually signs in at a terminal, then flips active. A plain Nurse assignment is a bedside/cart role and does not fill any character seat.

## NETWORK & TABLETS

### A tablet says the portal is unreachable — how do I get it connected?

Work through these in order: (1) You launched with the LAN launcher — start.sh, not run_cards.sh (run_cards binds loopback only, so tablets can't reach it). (2) The tablet is on the SAME Wi-Fi as the Mac. (3) On the tablet, Private Wi-Fi Address is OFF for this network (Settings → Wi-Fi → ⓘ → Private Wi-Fi Address → Off, then rejoin) — a randomized MAC gets silently dropped by the kit router. (4) The CA is trusted on the tablet (one-time onboarding). Run bash scripts/preflight.sh (add the tablet's IP to ping-test it); it flags exactly which of these is wrong.

### The Mac operator works fine but EVERY tablet fails (no image, no audio, "Send failed") — what's wrong?

That signature is IP-SAN drift: the TLS cert no longer covers the current LAN IP (DHCP moved the Mac, or the IP was minted as a name instead of a real IP). The page taps through the cert warning but the avatar image, speech, and device sends are subresources with no tap-through, so they're silently blocked. Fix: re-mint the leaf cert with the bare command — python scripts/dev_cert.py (it auto-detects the IP; do NOT pass the IP as a host argument, and NEVER use --remint) — then RESTART the portal so it loads the new cert, and re-scan the tablet QR. Verify with preflight.sh (all green) before handing anyone a QR. Newer builds auto-re-mint on launch, so mid-session this may be the only manual case.

### The animated patient face loads on the tablet but stays a generic demo face (not the patient's portrait) and PTT says "portal unreachable" — why?

This is the cross-origin (origin-mismatch) trap, not a cert problem: the portal wasn't serving the avatar app, so the QR sent the tablet to the separate dev server on :5173 while the face binding lives on the API at :8760 — the binding fetch is cross-origin and silently fails, so the rig never skins and push-to-talk can't reach the turn API. Fix: launch so the portal serves the avatar app from one origin — this is what run_cards.sh and start.sh do by default (VRAI_FACES_SERVE=portal). After relaunching, re-print/re-scan the QR so it points at :8760/face..., not :5173. Re-minting the cert or re-trusting the CA does NOT fix this — it's an origin issue.

### How do I onboard a brand-new tablet (iPad or Android) so it trusts the certificate?

One time per tablet: on the new device open the onboarding helper at http://<mac-ip>:8766 (plain HTTP, so no certificate chicken-and-egg) — the portal banner and preflight.sh print this URL. It serves per-platform steps and the CA download. On iPad: install the downloaded profile (Settings → Profile Downloaded → Install), then Settings → General → About → Certificate Trust Settings → toggle MedSim Dev Local CA ON, and set Private Wi-Fi Address OFF for the dev network. On Android/Chrome: Settings → search "CA certificate" → Install a certificate → CA certificate → pick the downloaded file. Then scan the QR. Done once, the device survives every future cert re-mint and router change.

### The tablet shows "This Connection Is Not Private" / a certificate warning — how do I clear it properly?

Don't just tap through it (a per-page exception dies with every cert re-mint and blocks the avatar/audio subresources anyway). The real fix depends on the cause: if the CA profile isn't installed or the trust toggle is off, finish onboarding — install the CA and toggle MedSim Dev Local CA ON under Certificate Trust Settings (iOS). If the CA is already trusted, the leaf cert likely doesn't cover the current IP: re-mint the leaf (python scripts/dev_cert.py) and restart the portal, then re-scan. Onboarding the CA once means future re-mints never re-warn.

### How do I print and use the station QR codes?

Open the QR sheet from Print QR on the Operate lifecycle bar (or the 🖨️ link from a bed / "Print" on the Multi-Patient join sheet). Each patient page leads with the 🪞 Patient avatar (animated rig) QR — scanning it on a bedside tablet loads the animated 3D face bound to that patient — followed by the 💬 Chat station, 📋 Medical Records Terminal, ⚕ Device station, and that bed's character avatar/voice QRs. A Common page carries shared characters and common devices (🩺 Nursing Station and the med cart), which aren't repeated per bed. After any network or IP change, re-print the sheet so codes carry the current IP.

### Can one tablet running a shared character (like the doctor) cover multiple beds?

Yes. Scan the single shared-character QR on one tablet, then push-to-talk or type about Bed 1 for a reply in that character's voice; on the same tablet ask about Bed 2 and it answers about the other patient — the shared doctor knows all the beds. The shared station supports Hold to talk (mic prompt → speech → spoken reply) and a type fallback.

### An iPad is showing an old version of a page after an update — how do I force the fresh build?

Do the close-open twice dance: fully swipe the Safari tab away, reopen and re-scan, then repeat once — this lets the service worker pick up the new build. If it's still stale, clear the site data: Settings → Safari → Advanced → Website Data → delete the entry for the Mac's LAN IP, then re-scan the QR. On the operator's Mac browser, a hard refresh (Cmd+Shift+R) drops a stale console.js.

**On the tablet, the first mic take is slow (~60 s) every single session — how do I fix it?**

You're almost certainly in a Private/Incognito tab, which has no persistent storage, so the ~70 MB speech model re-downloads every session. Use a normal (non-private) tab, or add the page to the Home Screen and open it as an app — the model then caches per origin and subsequent sessions warm quickly.

**Nothing loads on the tablet and the Mac can't even ping it — what now?**

The router is dropping device-to-device traffic. Two common causes: a randomized/rotating MAC (turn Private Wi-Fi Address OFF for this network on the tablet and rejoin), or the access point's client/AP isolation setting (disable it on the router). Note that a phone hotspot can be carrier IPv6-only (the Mac shows an IP like 192.0.0.2) which can't pair at all — preflight.sh flags that. Confirm both devices are on the same Wi-Fi.

**When I re-mint a certificate, do I have to re-trust every tablet again?**

No — as long as you only re-mint the LEAF cert, not the CA. The model is: one CA forever (each tablet trusts it once), disposable leaf certs. A new router or Mac IP just means re-minting the leaf (python scripts/dev_cert.py) and restarting the portal; the tablets trust it automatically because it chains to the CA they already have — no profile, no toggles. NEVER use --remint / REMINT_CA: that destroys the CA and forces re-trusting every device in the fleet.

## DEBRIEF & HANDOFF

**How do I end the session and produce the debrief?**

On the Operate tab's room lifecycle bar, click "End all (debrief)" and confirm. It builds the cohort debrief and opens it (one entry per bed's session), and any connected students are disconnected. Switch to the Debrief tab any time to reach "Open debriefs," and the cohort debrief lists each bed's session so you can drill into a per-patient debrief.

**Where do I review the debrief after a session?**

Go to the Debrief tab and open "Open debriefs," or use the Cohort debrief link. The cohort debrief lists every bed's session for review, and each one links through to a per-patient debrief. The cohort debrief is built automatically when you use "End all (debrief)" on the Operate lifecycle bar.

**How does the shift-handoff exercise and its SBAR score work?**

In the encounter console, use the Shift handoff card: Start it (choose the mode and counterpart), then engage the counterpart via push-to-talk to give or receive report. Click Score and you get the SBAR coverage grid (check/cross for each SBAR element) plus a self-versus-measured percentage, so trainees see what they covered and missed.

**How do I see which students are actually connected during a session?**

In the encounter console, the Connected stations card shows the live per-bed student roster — who's online, their platform, and turn counts. Instructor Engage stations are filtered out so they don't clutter the roster. On Operate → Network & devices you also see each rostered role's node and whether it's idle (rostered but not signed in) or active (signed in at a terminal).

**I drilled into a patient's encounter console — how do I get back?**

Click "← Back to Mission Control" at the top of the encounter console; it returns you to the Operate tab (not the classic room). Individual cards inside the console also collapse via their caret/header and can Pop out to their own live window.

## TROUBLESHOOTING

**A tablet shows a white screen with the URL in the bar and a spinner forever — what's wrong?**

That's TLS failing while a stale tab masks the certificate warning. Clear Safari's Website Data for the portal origin on that tablet and retry, then fix the underlying cert (make sure the CA is trusted and the leaf covers the current IP, re-minting and restarting if needed). Note the portal access log shows nothing for a cert-rejected handshake, so "zero log lines" does not mean "no traffic" — suspect TLS/trust first.

**The portal only loads in a Private/Incognito tab on the tablet — why?**

Normal-tab state (a stale per-cert exception or cached shell) is the culprit — this was confirmed not to be Private Relay. Clear Website Data for the portal origin and finish CA onboarding (install and fully trust the CA), then it works in a normal tab. Prefer a normal tab anyway, because Private tabs also re-download the speech model every session (slow first mic take).

**The tablet UI and panels load but the patient's face is a black void — what does that mean?**

That's the "ghost app": the offline shell booted from the service-worker cache, but the (uncached) face-binding API couldn't reach the portal — so the network/TLS is broken underneath. Diagnose the cert/network (leaf covers the current IP, CA trusted, portal serving one-origin from :8760), fix it, then re-scan the QR. Don't chase it as a rendering bug; it's a connectivity failure showing through the cache.

**I get a 403 / "sign in with the Admin seat" when opening Credentials — why?**

The Credentials page and EHR admin are Admin-only. If you're signed in as Instructor or Observer you'll get a 403 with a message to use the Admin seat. Sign out and sign back in with the master password (the Admin seat) — the page then opens and shows the Seat passwords section, where you can set optional Instructor/Observer passwords (8-character minimum).

**A character turn errors with "No Anthropic API key configured" — how do I resolve it?**

No AI key is set (or it was cleared). As Admin, open the Credentials page, paste the Anthropic key, click Save, then Test to confirm it's accepted. The next push-to-talk or typed turn picks up the key without restarting the scenario. If a student hits it, they'll see "ask your instructor to add it" — that's the instructor's cue to set the key on Credentials.

**When something fails during a field test, what info should I capture to report it?**

Note the failing step number, what you saw versus what you expected, the browser, and whether you hard-refreshed (Cmd+Shift+R drops a stale console.js on the Mac; on iPad do the close-open twice dance). Include any console error or the portal log at /tmp/medsim_portal.log. For speech- or bundle-related issues, also state whether the tablet was on the fresh build and which launcher you used (run_cards vs start.sh). That's exactly what's needed to pin the fix to the step.

## EVIDENCE & REFERENCES

**Is the debriefing and ranking based on real educational science, or is it made up?**

It is grounded in established healthcare-simulation science, not invented. The debrief follows the PEARLS framework (Reactions, Description, Analysis, Application, Summary); the facilitator stance uses 'debriefing with good judgment' (advocacy-inquiry); the handoff is scored on SBAR; and the per-encounter 'clinical-judgment coverage' maps to Tanner's Clinical Judgment Model and the NCSBN Clinical Judgment Measurement Model (the model behind the Next Generation NCLEX). See Part III of the Operator Guide for the full map and references.

**What exactly does the performance ranking measure, and against what standard?**

The ranking is a structured, criterion-referenced judgement — not a hidden score. It reflects clinical-judgment coverage (noticing, interpreting, responding, reflecting — Tanner's model), SBAR handoff coverage, and engagement, summarized per encounter in the debrief. Simulation assesses the 'shows how / does' tiers of Miller's pyramid. The criteria are visible in the debrief and editable by the instructor, so you can align them to the Lasater Clinical Judgment Rubric, the Creighton C-SEI/C-CEI, or your program's QSEN competencies.

**Where is the list of references and research the system is based on?**

Part III of the Operator Guide ('The evidence base — standards & science') contains the curated reference list, grouped by learning science, debriefing, clinical-judgment & competency assessment, and standards/safety. It cites the foundational works (Kolb; Rudolph et al.; PEARLS; Tanner; INACSL Standards; NCSBN NCJMM; Lasater; Miller; and others). Instructors and students can look these up to see how the system is grounded.

### Which professional simulation standards does the system follow?

It is built to sit inside the INACSL Healthcare Simulation Standards of Best Practice (design, facilitation, debriefing, evaluation, professional integrity) and the NLN/Jeffries Simulation Theory, and to support QSEN safety competencies and SBAR communication. This lets a program defend the tool's use inside a nursing or allied-health curriculum.

### We want to run a research study — how can we build on or modify the system?

Every run is debriefed against a visible, editable rubric and recorded, producing a structured, timestamped dataset (per-encounter performance frames, SBAR coverage, clinical-judgment coverage, engagement, and student commitments). Good directions: validate the AI-assisted clinical-judgment coverage against a human-scored Lasater CJR or C-SEI/C-CEI; relate debrief measures to NGN/NCJMM readiness; use DASH to study facilitation; or test deliberate-practice dosing. Define your construct and instrument, configure the rubric to match, export the data, and pre-register. Part III lists the anchor references. The rubric and scenario library are meant to be modified for a study.

### What is 'clinical-judgment coverage' in the debrief analysis?

It reflects Tanner's Clinical Judgment Model — did the learner notice the relevant cues, interpret them correctly, respond appropriately, and reflect. This is the same construct the NCSBN Clinical Judgment Measurement Model uses for the Next Generation NCLEX, so the debrief speaks the language of current licensure assessment. It is reported per encounter and can be aligned to a validated rubric.

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